TEACHER GUIDE DRAFT

IB Diploma Programme Business management HL

Stock

Unit 5: Operations management
5.6 Production planning (HL only)



A revision lesson. Pupils run a car dealership in firms of four and revise stock on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min	45 min	60 min	75 min

What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal stock

Topic tasks

Each firm gets its own figures. The method is the same.

1 Stock holding cost

Round 1

Cars wait in your yard.

Cost of holding the cars?

You hold for them	**10 cars**
Holding cost per car a week	**£90**
Weeks you hold them	**6 weeks**

ANSWER

£5,400

10 × £90 × 6 = £5,400

Why: Stock that waits costs £900 every week.

2 Risks of JIT

Round 1

You move to just-in-time stock.

Which two risks rise most?

- **A** More cars go out of date
- **B** Higher storage costs
- **C** A late delivery stops sales
- **D** Fewer bulk-buying discounts
- **E** More cash tied up in stock

ANSWER

✓ A late delivery stops sales
✓ Fewer bulk-buying discounts

Why: JIT frees cash and space, but depends on every delivery and buys in small lots.

3 Rank stock turnover

Round 2

Three dealers: cost of cars sold in a year, and average stock.

Order by stock turnover, fastest first?

- **A** £450,000 sold, £50,000 stock
- **B** £800,000 sold, £200,000 stock
- **C** £600,000 sold, £100,000 stock

ANSWER

1. **£450,000 sold, £50,000 stock**
2. **£600,000 sold, £100,000 stock**
3. **£800,000 sold, £200,000 stock**

Why: Turnover: 9, 6 and 4 times a year. The biggest dealer is the slowest.

Topic tasks

4 Stock turnover

Round 2

One model, one year.

Stock turnover, times a year? Use average stock.

Cost of cars sold in a year	**£167,400**
Stock at start of year	**£15,420**
Stock at end of year	**£21,780**

ANSWER

9 times

(£15,420 + £21,780) ÷ 2 = £18,600; £167,400 ÷ £18,600 = 9

Why: The stock is sold and replaced 9 times a year.

5 Cut or raise buffer

Round 3

Deliveries now come a week late. A lost sale costs £1,500; one more car in stock costs £80 a week.

What should you do with buffer stock, and why?

A Cut it to save cash

B Raise it

ANSWER

✔ **Raise it**

Why: A lost sale costs far more than a week's holding cost, so a bigger buffer pays.

6 Weeks until reorder

Round 3

When do you order next?

Weeks until you reorder?

They buy each week	**4 cars**
Supplier delivers in	**2 weeks**
Buffer, of lead-time sales	**50%**
In stock	**28 cars**

ANSWER

£4

Lead-time sales 8, buffer 4, level 12; (28 − 12) ÷ 4 = 4

Why: A buffer tied to lead time grows when deliveries are slow.

Final questions

Each pupil answers 6 final questions on stock. They come from these 8 questions.

1 What is the central trade-off when a dealership adopts just-in-time delivery for common parts?

✔ **Lower holding cost, more supply risk**

2 A critical component supplier fails with no warning. Which earlier policy would most likely have reduced impact without necessarily holding maximum stock?

✔ **Tested backup suppliers for critical parts**

3 The cheapest supplier delivers only 72% of orders on time and in full, causing cancellations. What should Operations do?

✔ **Weigh failure cost against price**

4 A workshop uses just-in-time parts supply. Its main supplier shuts for two weeks after a fire. What is the most likely effect?

✔ **No buffer, so jobs stop at once**

5 Your only parts supplier's lead time rises from two days to nine. Apart from ordering earlier, what else must change?

✔ **More stock must be held, tying up cash and space**

6 A workshop holds 360 oil filters and uses 24 a day. How many days of stock does it hold?

✔ **15 days**

7 Average stock is worth £120,000 and holding costs run at 15% of stock value a year. What is the annual holding cost?

✔ **£18,000**

8 A dealer moves to just-in-time parts delivery. Which risk rises?

✔ **A late delivery stops repairs**